

MA 312: Design and Analysis of Algorithms

Course Code	MA 312
Course Name	Design and Analysis of Algorithms
L-T-P-C	3-0-2-4
Prerequisites	IC253
Intended for	B. Tech Maths and Computing
Distribution	Core
Mutual Exclusion	CS212, CS403
Approval	59th BoA (Apr 2025)

Course Contents

- **Unit I:** The Role of Algorithms in Computing, Analyzing Algorithms, Best, Average, Worst Case Analysis, Asymptotic notations, big-oh, theta, big-omega, little-oh and little-omega, Recurrences, Master's Theorem, Amortized analysis: aggregate analysis, accounting, potential method. [7 Lectures]
- **Divide and Conquer:** Merge Sort, Quick Sort, Finding Closest Pair of Points, Strassen's Algorithms, Median Finding, Discrete Fourier Transforms. [7 Lectures]
- **Greedy Algorithms:** Un-weighted Interval Scheduling, MST-- Prims and Kruskals, Dijkstra's shortest path, fractional knapsack, and Huffman coding. [7 Lectures]
- **Dynamic Programming:** Integral knapsack, Weighted Interval Scheduling, longest common subsequence, Rod-Cutting, matrix chain multiplication, and independent sets in trees. [7 Lectures]
- **Geometric Algorithms:** Line Segments: Properties, Intersection, sweeping lines- Convex Hull finding algorithms. [4 Lectures]
- **Intractability:** NP-completeness: reduction amongst problems, classes NP, P, NP-complete, and polynomial time reductions. [4 Lectures]
- **Introduction to Advanced techniques:** Approximation Algorithms, Vertex Cover, Set Cover, Minimum Makespan, Subset Sum Problem. Parameterized Algorithms, Vertex Cover, Feedback Vertex Set in Tournaments. [6 Lectures]

Laboratory Modules

Coding Lab will cover topics discussed in this course.

Text Books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, **Introduction to Algorithms**, MIT Press, 3rd Edition, 2009.
2. J. Kleinberg and E. Tardos, **Algorithm Design**, Pearson, 2006.

References

1. Dasgupta, Papadimitriou and Vazirani, **Algorithms**, McGraw-Hill Education, 2006.
2. Horowitz, Sahni, and Rajasekaran, **Computer Algorithms**, Silicon Press, 2007.